

NLP Mid Exam 2025/2026

1. Which of the following best distinguishes lemmatization from stemming in NLP?
 - A. Lemmatization uses predefined suffix rules, while stemming depends on vocabulary lookup.
 - B. Stemming uses context-based transformations, while lemmatization simply truncates words.
 - C. **Lemmatization preserves the grammatical and morphological analysis, whereas stemming does not.**
 - D. Stemming always produces valid dictionary words, while lemmatization may not.
2. Why is the Continuous Bag-of-Words (CBOW) model considered faster but less effective for rare words compared to Skip-gram?
 - A. It ignores the target word and predicts its surrounding context words.
 - B. **CBOW averages multiple context words to predict one target, reducing computational complexity but losing rare word distinctions.**
 - C. CBOW uses a higher-dimensional hidden layer that overfits frequent words.
 - D. CBOW requires more training for rare words, increasing noise.
3. In Word2Vec training, what happens when the window size parameter is increased?
 - A. It reduces the number of training samples generated.
 - B. **It captures a broader range of contextual words, thus capturing broader semantic relationships.**
 - C. It increases overfitting by limiting context.
 - D. It prevents the model from learning syntactic relations.
4. What is a fundamental computational drawback of Bag-of-Words (BoW) and One-Hot Encoding that Word Embeddings overcome?
 - A. They cannot handle non-English vocabularies.
 - B. **They produce high-dimensional sparse vectors that ignore semantic relationships between words.**
 - C. They are labeled data for training word representations and rely on deep neural networks for vectorization.
 - D. They introduce biases from word frequency.
5. In a Recurrent Neural Network (RNN), what is the primary role of the feedback loop?
 - A. To propagate data forward only.
 - B. **To enable the network to remember previous inputs.**
 - C. To increase the number of output neurons.
 - D. To reduce the number of input connections.
6. When unrolling an RNN through time, what is a key effect on the network structure?
 - A. **It converts the RNN into a feedforward network with multiple inputs and outputs.**
 - B. It removes the biases from the network.
 - C. It consolidates multiple layers into one layer.
 - D. It eliminates the need for activation functions.
7. Which of the following best describes the purpose of Word Embeddings in NLP?
 - A. To represent text using word frequency counts only.
 - B. **To convert words into numeric vectors that capture semantic and syntactic meaning.**
 - C. To create sparse matrices for text classification.
 - D. To store each word as a unique binary index.

8. What is a major limitation of the Bag-of-Words (BoW) model?

- A. It captures the semantic relationships between words.
- B. It requires labeled data for training.
- C. It ignores the order and context of words in a document.
- D. It reduces dimensionality automatically.

9. Which of the following best describes the difference between RNNs and standard feedforward neural networks?

- A. RNNs include feedback loops that enable temporal dynamic behavior.
- B. RNNs do not use activation functions.
- C. Feedforward networks are unrolled across time unlike RNNs.
- D. RNNs do not have weights and biases.

10. In the Skip-gram model, what serves as the input and what is predicted as output?

- A. Input: context words, Output: target word.
- B. Input: document, Output: topic.
- C. Input: target word, Output: context words.
- D. Input: frequency count, Output: embedding vector.

NLP Quiz 2025/2026

1. What is the main idea behind Word2Vec?

- A. Convert words into one-hot vectors
- B. Convert words into dense vectors based on context
- C. Convert sentences into images
- D. Count the frequency of words

2. Which of the following is NOT a training architecture of Word2Vec?

- A. Skip-Gram
- B. CBOW
- C. GloVe
- D. Both A and B

3. What is a unique advantage of FastText over Word2Vec?

- A. It trains faster
- B. It represents words as bags of characters (sub-word units)
- C. It produces larger embeddings
- D. It only works on English

4. Why does FastText handle rare words better than Word2Vec?

- A. It ignores rare words
- B. It uses subword information
- C. It increases embedding size
- D. It uses more epochs

5. What problem do RNNs commonly suffer from?

- A. Overfitting
- B. Vanishing and exploding gradients
- C. Slow inference
- D. Memory overflow

6. Which mechanism allows LSTMs to overcome the vanishing gradient problem?

- A. Large batch size
- B. One-hot encoding
- C. **Gated memory cells (forget, input, output gates)**
- D. More layers

7. What is the main advantage of LSTM over a simple RNN?

- A. It trains faster
- B. **It stores long-term dependencies better**
- C. It reduces dataset size
- D. It never overfits

8. Which of the following is true about RNNs?

- A. They process all inputs independently
- B. **They maintain a hidden state that carries information across time**
- C. They can only process images
- D. They don't use activation functions

9. In Word2Vec Skip-Gram, the model predicts:

- A. The word from its characters
- B. **The context words given the target word**
- C. The next sentence
- D. The next character

10. FastText is particularly useful for:

- A. **Languages with complex morphology**
- B. Short English sentences only
- C. Tasks requiring image embeddings
- D. Noise reduction in audio